Review Count and Balancing Summary

Threshold 0.3

1. Review Count Effect: No Balancing Method

Corpus	Rows	Perfumes	LR F1	XGBoost F1
Synthetic 784x3	99	33	Mean 0.5931 SD 0.0491	Mean 0.5830 SD 0.0906
	201	67	Mean 0.5998 SD 0.0276	Mean 0.5877 SD 0.0533
	300	100	Mean 0.6043 SD 0.0141	Mean 0.6087 SD 0.0202
	501	167	Mean 0.6320 SD 0.0389	Mean 0.6055 SD 0.0140
	1002	334	Mean 0.6385 SD 0.0142	Mean 0.6251 SD 0.0157
	2001	667	Mean 0.6307 SD 0.0072	Mean 0.6318 SD 0.0095
	2352	784	**Mean 0.6514** SD 0.0000	**Mean 0.6576** SD 0.0056
Synthetic 784x6	198	33	Mean 0.5524 SD 0.0622	Mean 0.5495 SD 0.0675
	402	67	Mean 0.5556 SD 0.0264	Mean 0.5494 SD 0.0173
	600	100	Mean 0.5744 SD 0.0167	Mean 0.5638 SD 0.0282
	1002	167	Mean 0.5790 SD 0.0112	Mean 0.5616 SD 0.0096
	2004	334	Mean 0.5874 SD 0.0079	Mean 0.5596 SD 0.0124
	4002	667	Mean 0.5829 SD 0.0054	Mean 0.5797 SD 0.0052
	4704	784	**Mean 0.5876** SD 0.0000	**Mean 0.5800** SD 0.0018
YouTube only	784	784	**0.3964**	**0.3601**
YouTube + synthetic 784x3	3136	784	**0.4131**	**0.3609**
YouTube + synthetic 784x6	5488	784	**0.3526**	**0.4171**

No balancing method; mixed rows are evaluated on YouTube rows.

Emotion-count balancing

Main comparison: no balancing, hybrid, and SMOTE.

2.0 Summary: No Balancing vs Hybrid vs SMOTE

Training data	Logistic Regression			XGBoost		
	No balancing	Hybrid	SMOTE	No balancing	Hybrid	SMOTE
Synthetic 784x3	**0.6514**	0.6022	0.6358	0.6576	0.6130	**0.6577**
Synthetic 784x6	**0.5876**	0.5456	0.5731	0.5800	0.5472	**0.5967**
YouTube only	**0.3964**	0.2735	0.3812	0.3601	0.2346	**0.3695**
YouTube + synthetic 784x3	**0.4131**	0.3323	0.3787	0.3609	**0.3684**	0.3445
YouTube + synthetic 784x6	0.3526	0.3483	**0.4051**	**0.4171**	0.4058	0.3662

Best F1 in each row/model is shaded.